

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
3	(a)	(i)	$1.79 \text{ eV} = 2.86 \times 10^{-19} \text{ J}$ or by implication (1) Use of $hf = E_U - E_L$ and $\lambda = \frac{c}{f}$, or $\lambda = \frac{hc}{E_U - E_L}$ or equiv (1) $\lambda = 694 [\pm 1] \text{ nm}$ / $6.9 \times 10^{-7} \text{ m}$ (1)	1	1 1		3	2	
		(ii)	So a photon is more likely to cause stimulated emission than to be absorbed / more stimulated emission than absorption (or so more photons cause SE than are absorbed) (1) So number of photons increases [rather than decreases] / light builds up / 1 photon $\rightarrow$ 2 photons (1)	2			2		
	(b)	(i)	$\text{photons per second} = \frac{6.0 \times 10^{-3} [\text{W}]}{2.86 \times 10^{-19} [\text{J}]}$ ecf even if slips (eg by 10^n) (1) $2.09 \times 10^{16} [\text{s}^{-1}]$ (1)	1	1		2	1	
		(ii)	$\text{photon momentum} = \frac{6.63 \times 10^{-34} [\text{J s}]}{693 \times 10^{-9} [\text{m}]} [= 9.55 \times 10^{-28} \text{ N s}]$ or by implication ecf on λ (1) or beam momentum per second = $\frac{0.006 [\text{W}]}{c}$ or by implication beam momentum per second = $2.0 \times 10^{-11} [\text{N}]$ (1) [$1.91 \times 10^{-11} \text{ N}$ if 2×10^{16} used]	1	1		2	2	
		(iii)	$4.0 \times 10^{-11} \text{ N}$ ((unit)) ecf from (ii), i.e. $2 \times$ answer to (ii) But not if answer to (ii) was 0.		1		1	1	
			Question 3 total	5	5	0	10	6	0